SMART INDIA HACKATHON 2026

- **Problem Statement ID – 26118**
- **Problem Statement Title-** Passive Colorimetric H2S Exposure-Dosimeter Wristband with AI-Based Quantitative Reading
- **Theme-** Smart Automation
- **PS Category-** Hardware
- **Team ID-**
- **Quantum Realm**

Quantum Realm

IDEA TITLE

SMART INDIA HACKATHON 2026

Proposed Solution

- **Upcycled Waste-to-Wealth Chemistry:** Utilizing industrial spent PCB etchant for low-cost sensing.
- **Whoop-Style Ergonomic Wearable:** Lightweight, comfortable elastic wristband housing.
- **Chevron Progress Bar:** 3 angled sensing tracks of varying thickness for rate-controlled gas diffusion.
- **Pre-Shift Expiry Indicator:** Built-in moisture-sensitive cell.
- **AI Computer Vision Scanning:** Smartphone app auto-calibrates against an integrated color reference scale.
- **Environmental Compensation:** Algorithmic dose adjustment based on real-time ambient temperature and humidity.
- **Automated Compliance Logging:** Digital recording of worker exposure dose (ppm·hours) tied to Employee ID for safety reporting.

Wearable Patch

A Whoop-style ergonomic wristband featuring a rate-controlled chevron patch and pre-shift expiry indicator for passive, battery-free exposure tracking.

Mobile App Dashboard

A mobile app dashboard displaying weekly exposure summaries, shift logs, and regulatory compliance status for DGMS and OISD reporting.

Spent Etchant Gel (Zero-extraction waste ink)

PDMS Diffusion Barrier (Controlled gas transport)

Chevron Progress UI (Visual exposure levels)

Integrated Reference Scale (On-badge color calibration)

OpenCV Lighting Engine (Shadow/glare normalization)

Temp/Humidity Adjustment (Algorithmic accuracy correction)

Expiry Silica Indicator (Pre-shift badge validity check)

DGMS/OISD Auto-Reports (One-click audit exports)

"From Waste to a Safer Tomorrow"

It Addresses the Problem by

- **Focusing on Cumulative Dose:** Tracks long-term exposure ("concentration" × "time") rather than short-lived instantaneous peak alarms.
- **Zero Maintenance & Battery-Free:** Eliminates charging, calibration, and electronic upkeep required by active gas detectors.
- **Tamper-Proof Validity:** Visual expiry indicator ensures workers never wear compromised or degraded badges.
- **Automated Regulatory Compliance:** Replaces manual pen-and-paper logs with digitized exposure histories meeting safety standards.

Android | iOS | Web

H_2S DoseVISION Platform

H_2S DOSEVISION

- Admin Analytics & Shift Heatmaps
- Worker ID Exposure Profiles
- DGMS / OISD Report Generator
- Site Leak Alerts & Safety Escalations

Quantum Realm

TECHNICAL APPROACH

SMART INDIA HACKATHON 2026

1. User & Admin Interfaces (Client Side)

Worker Mobile App UI

Modules
- Pre-Shift Check-In
- Scan Viewfinder (OpenCV Auto-Crop)
- Shift Summary
- Exposure History
- Compliance Status

Actions
- Camera capture
- QR / Employee ID authentication
- Local storage caching

Modules
- Admin Dashboard
- Live Site Heatmaps
- Cumulative Exposure Statistics
- DGMS/OISD Auto-Report Generator

Actions
- Export PDF audit logs
- Manage worker safety rotations
- Monitor multi-sector gas alerts

2. Core Client Apps, Hardware & Deployment

Mobile App & Wearable Hardware

Whoop-style wristband with rate-controlled chevron patch

Scan via Smartphone (Flutter / OpenCV)

- Image Processing (OpenCV)
- Lighting Normalization (using reference scale)
- Local Storage Cache (on device)

Secure Data Sync

Website Portal (Admin)

Tech Stack
- React.js / Next.js
- Tailwind CSS
- TypeScript
- Chart.js / Recharts

CI/CD Deployment

Git → Docker → GitHub Actions → Cloud Deployment

3. Backend Server, Cloud & External APIs

Backend Cache

Stores cached weather parameters and recent scan data to reduce redundant API calls.

Backend Server (Node.js / Express / Python FastAPI)

- Processes image packets
- Executes dose calculation algorithms (factoring temp/humidity)
- Enforces rate limits
- Utilizes external APIs

External APIs

Weather API

Real-time ambient temperature & humidity correction factors.

OpenCV / AI Engine

Color-to-dose conversion mapping.

Cloud Storage & Database

PostgreSQL AWS S3

- Stores user ID records, shift logs, cumulative exposure history
- Stores uploaded badge scan images for compliance verification
- Stores official report templates

Quantum Realm

FEASIBILITY AND VIABILITY

SMART INDIA HACKATHON 2026

Financials

Revenue Model

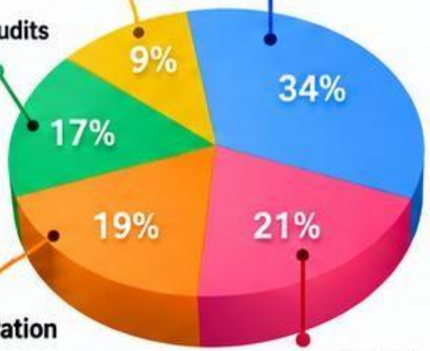
- B2B industrial safety subscriptions
- Bulk patch contracts with energy/mining enterprises
- Premium analytics dashboards

Cost Allocation

Operations
9%
Foil-pouch packaging and logistics

Patch Manufacturing
34%
Upcycled etchant hydrogel, PDMS casting, and PET substrates

Compliance & Audits
17%
DGMS/OISD documentation management



R&D & Calibration
19%
Arduino chamber testing and algorithm tuning

Cloud & App
21%
Flutter app maintenance, OpenCV hosting, and database management

Feasibility

Zero-Extraction Upcycling
Direct use of spent PCB etchant eliminates chemical processing and high CapEx.

Rate-Controlled Diffusion
Variable PDMS membrane thicknesses (0.2 mm to 1.0 mm) create a reliable visual progress bar.

Offline-First App
Local caching ensures seamless barcode scanning and shift logging in remote areas with weak connectivity.

Lighting Normalization
On-badge color reference scales allow OpenCV to instantly correct harsh industrial shadows or glare.

Viability

High Market Demand
Aligns with mandatory worker safety regulations enforced by DGMS and OISD.

Cost Disruption
Battery-free, disposable patches drastically undercut expensive electronic gas detectors.

Sustainable Supply
Industrial partnerships secure free active material while solving e-waste disposal burdens.

Digital Transformation
Replaces manual logs with centralized cloud dashboards and automated audit reports.

Challenges and Mitigation

Challenge	Mitigation
Environmental Skew Extreme heat or humidity alters reaction speeds.	Real-time algorithmic correction via weather APIs adjusts final dose calculations.
Storage Degradation Moisture breaching badges prior to field use.	Airtight foil packaging paired with a moisture-sensitive expiry dot for pre-shift validation.
Device Fragmentation Varying smartphone camera sensors across worker devices.	Automated bounding-box detection referencing the fixed scale printed on the patch border.
Regulatory Approval Meeting strict safety board standards.	Output formats built natively to satisfy DGMS and OISD compliance requirements.

Quantum Realm

IMPACT AND BENEFITS

SMART INDIA HACKATHON 2026

IMPACT

Worker Health & Safety

Continuous cumulative dose tracking (ppm·hours) protects industrial workforces from chronic H_2S exposure before symptoms arise.

Cost Disruption

Battery-free, disposable Whoop-style patches eliminate the charging, calibration, and high maintenance costs of traditional electronic detectors.

Operational Efficiency

Automated digital compliance logging and cloud synchronization replace manual pen-and-paper records, streamlining safety audits.

Environmental Sustainability

Upcycling hazardous spent PCB etchant into a reactive hydrogel supports circular-economy and green manufacturing goals.

BENEFITS

Industrial Workers

Lightweight, ergonomic wristband design ensures zero friction, high comfort, and reliable pre-shift safety verification.

Safety Officers & Management

Centralized web dashboards, heatmaps, and automated DGMS/OISD reports enable proactive leak detection and staff rotation.

Plant Operations

Zero-maintenance, offline-first design functions seamlessly in remote extraction and mining sectors without requiring external power.

Regulatory Bodies

Standardized audit trails and tamper-proof validity checks ensure total adherence to national industrial health mandates.

Quantum Realm

RESEARCH AND REFERENCES

SMART INDIA HACKATHON 2026

Research Papers & Standards

Active Chemical Sensing
Colorimetric Detection of Hydrogen Sulfide via Metal-Ligand Interactions
Smith et al., ACS Applied Materials, 2023

Diffusion Mechanics
Rate-Controlled Gas Transport Across Porous PDMS Membranes
Journal of Membrane Science, 2024

Upcycled E-Waste
Valorization of Spent Cupric Chloride Etchant in Functional Hydrogels
Elsevier, 2025

Industrial Safety Frameworks
DGMS Guidelines on Occupational Exposure Limits for Toxic Gases in Mines
Directorate General of Mines Safety, 2023

Standard Regulations
OISD Standard-194: Storage, Handling and Safety Rules for Hydrogen Sulfide Environments
Oil Industry Safety Directorate

Competitors Comparison

Features (Criteria)	H_2S-DoseVision	Electronic Gas Detectors	Lead-Acetate Badges	Smart Watch Monitors
Cumulative Dose (ppm · hours)	✓	✗	✗	✗
Battery-Free / Passive	✓	✗	✓	✗
Upcycled Materials (Waste-to-Wealth)	✓	✗	✗	✗
Pre-Shift Expiry Indicator	✓	✗	✗	✗
AI Computer Vision Scanning	✓	✗	✗	✓
DGMS/OISD Auto-Reporting	✓	✗	✗	✗
Unit Cost	✓ (Very Low)	✗ (Very High)	✓ (Low)	✗ (High)

Industry Outlook & Statistics

$1.8B	Global Industrial Gas Detection Market by 2030 (CAGR ~8.4%)
₹12,000 Cr	Indian Oil & Gas PPE and Safety Compliance Market
100%	Shift from instantaneous alarms to cumulative dose tracking required for modern OSH standards
10x	Cost reduction of disposable passive dosimeters versus active electronic monitors
Zero	External power or battery requirements for continuous wear

Primary Research (Field Survey Insights)

78.4%	of surveyed industrial workers reported that current electronic monitors are too heavy or cumbersome for full-shift comfort.
64.2%	stated that maintenance and battery charging are frequently neglected on-site.
52.1%	highlighted a lack of reliable tracking for cumulative low-level chronic exposure during regular shifts.
41.3%	confirmed that pen-and-paper compliance logbooks are prone to human error or lagging entries.